

File 31: UGA Intern Workstream — Page 4: USACE Savannah District 7-Year Monitoring & Release Schedule

Ecology & Biological Monitoring Training Manual Target School: University of Georgia (UGA) Odum School of Ecology / Warnell School of Forestry Project Area: Upper Savannah Basin & Chattahoochee River Spawning Headwaters Supervising Board: Hunter Morris (Co-Founder & Managing Director) & Hadi Irvani (Board Member)

1. Regulatory Context: Credit Release Milestones

A mitigation bank does not receive all of its commercial stream credits at once. The **USACE Savannah District** releases credits in stages over a **7-year monitoring window**. These releases are strictly contingent upon demonstrating physical channel stability and biological success. As a UGA Intern, you will compile the data from your biological, thermal, and vegetative surveys into our annual **Ecological Monitoring Reports**. The USACE will not release credit reserves if we fail to submit these reports on time, or if our biological metrics fail to meet target thresholds.

2. The 7-Year Stream Credit Release Schedule

Our flagship bank at Anderson Creek yields exactly **12,900 credits**. The phased release schedule and required biological performance thresholds are detailed below:

Year	Released %	Released Credits	Critical Ecological & Geomorphic Performance Milestones
Year 0	15%	1,935 credits	Baseline: Mitigation Banking Instrument (MBI) approved; conservation easement recorded; credit security bonds posted.
Year 1	15%	1,935 credits	As-built survey submitted; bank stability index (BEHI) rated "Good" or better; vegetative survival $\geq 90\%$.
Year 2	15%	1,935 credits	Zero stream bank failure after bankfull storm event; vegetative density $\geq 260 \text{ stems/acre}$; invasive coverage $< 10\%$.

Year	Released %	Released Credits	Critical Ecological & Geomorphic Performance Milestones
Year 3	\$10\%\$	1,290 credits	Stream temperature loggers prove max summer temperature $<65^{\circ}\text{F}$; benthic macroinvertebrate EPT index increases by lge 15\%.
Year 4	\$10\%\$	1,290 credits	Rosgen cross-section width-to-depth ratios stable ($\pm 10\%$ of as-built designs); native vegetation canopy covers lge 40\% of plot surface.
Year 5	\$15\%\$	1,935 credits	Benthic EPT index matching reference stream conditions; native vegetative density stable; invasive coverage $<5\%$.
Year 6	\$10\%\$	1,290 credits	Multi-year thermal logging proves consistent temperature suppression; biological community shows high ecological lift.
Year 7	\$10\%\$	1,290 credits	Final Release: Complete geomorphic stability verified; permanent stewardship transferred to Georgia Land Trust.

3. Calculating the Benthic EPT Index

To verify biological recovery at Year 3 and Year 5, you must calculate the **EPT (Ephemeroptera, Plecoptera, Trichoptera) Index**. EPT organisms are highly sensitive to siltation and organic pollutants; their abundance is direct evidence of water quality improvement.

$$\text{EPT Index} = \frac{\text{Taxa Count of (Mayflies + Stoneflies + Caddisflies)}}{\text{Total Taxa Count of All Collected Macroinvertebrates}} \times 100\%$$

Worked Example:

You collect a sample at Anderson Creek Station 02 containing the following macroinvertebrates:

- **Ephemeroptera (Mayflies):** 4 distinct taxa (Heptageniidae, Baetidae, Ephemerellidae, Siphonuridae)
- **Plecoptera (Stoneflies):** 2 distinct taxa (Perlidae, Pteronarcyidae)
- **Trichoptera (Caddisflies):** 3 distinct taxa (Hydropsychidae, Limnephilidae, Rhyacophilidae)

- **Chironomidae (Non-biting Midges - Pollution Tolerant):** 5 distinct taxa
- **Odonata (Dragonflies - Moderately Tolerant):** 2 distinct taxa

1. Calculate EPT Taxa:

$$\text{EPT Taxa} = 4 + 2 + 3 = 9 \text{ taxa}$$

2. Calculate Total Taxa:

$$\text{Total Taxa} = 9 + 5 + 2 = 16 \text{ taxa}$$

3. Calculate EPT Index:

$$\text{EPT Index} = \frac{9}{16} \times 100\% = 56.25\%$$

Ecological Assessment: An EPT Index **>50%** represents an **"Excellent"** water quality rating, satisfying the USACE Year 3 ecological lift requirement.

4. USACE Annual Submission Checklist

Before submitting the annual report to the Savannah District IRT, you must verify that the following files are compiled and attached:

- [] **Executive Summary Sheet:** Formatted with Co-Founder Hunter Morris's signature and Board Member Hadi Irvani's review stamp.
- [] **HOBO Logger Data Plots:** Continuous summer water temperature telemetry graphs with critical 65°F line highlighted.
- [] **Vegetation Plot Sheets:** Native stem counts, diameter measurements, and calculated stems-per-acre density for all 5 monitoring plots.
- [] **Macroinvertebrate Taxa Log:** Complete species list, calculated EPT indices, and photographic voucher plates of identified specimens.
- [] **Geomorphic Cross-Sections:** Overlay plots comparing pre-construction, as-built, and current year channel cross-sections to prove dimensional stability.
- [] **Ground Photographic Plates:** Fixed-point photos facing upstream, downstream, and perpendicular at every cross-section monument.